

Product Review and Rating Management System

Lenskart – Eyewear E-Commerce Database System

1. Create Review and Rating Tables

1.1 Review

Column	Type	Constraint
Review_ID	NUMBER	Primary Key
Customer_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Review_Text	VARCHAR2(500)	—
Review_Date	DATE	—

1.2 Rating

Column	Type	Constraint
Rating_ID	NUMBER	Primary Key
Review_ID	NUMBER	Foreign Key
Rating	NUMBER	—

2. Review Table

```

SQL> CREATE TABLE Review(
  2     Review_ID NUMBER PRIMARY KEY,
  3     Customer_ID NUMBER,
  4     Product_ID NUMBER,
  5     Review_Text VARCHAR2(500),
  6     Review_Date DATE,
  7     FOREIGN KEY (Customer_ID) REFERENCES LenskartCustomer(Customer_ID),
  8     FOREIGN KEY (Product_ID) REFERENCES LenskartProduct(Product_ID)
  9 );

Table created.

```

3. Rating Table

```

SQL> CREATE TABLE Rating(
  2     Rating_ID NUMBER PRIMARY KEY,
  3     Review_ID NUMBER,
  4     Rating NUMBER(1) CHECK (Rating BETWEEN 1 AND 5),
  5     FOREIGN KEY (Review_ID) REFERENCES Review(Review_ID)
  6 );

Table created.

```

4. Review Data Insertion

```

SQL> INSERT INTO Review VALUES (601, 1, 1, 'Good quality frame and comfortable to wear',TO_DATE('24-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Review VALUES (602, 2, 2, 'Very useful for computer and laptop use',TO_DATE('25-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Review VALUES (603, 3, 3, 'Stylish design and good fitting',TO_DATE('26-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Review VALUES (604, 4, 4, 'Good sunglasses with excellent look',TO_DATE('27-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO Review VALUES (605, 5, 5, 'Comfortable contact lenses and easy to use',TO_DATE('28-09-2026','DD-MM-YYYY'));
1 row created.

```

4.1 Display Review Records

```

SQL> SELECT*FROM Review;

  REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      601           1           1
Good quality frame and comfortable to wear
24-SEP-26

      602           2           2
Very useful for computer and laptop use
25-SEP-26

  REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----

      603           3           3
Stylish design and good fitting
26-SEP-26

      604           4           4
Good sunglasses with excellent look

  REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
27-SEP-26

      605           5           5
Comfortable contact lenses and easy to use

```

5. Rating Data Insertion

```

SQL> INSERT INTO Rating VALUES (701, 601, 5);
1 row created.

SQL> INSERT INTO Rating VALUES (702, 602, 4);
1 row created.

SQL> INSERT INTO Rating VALUES (703, 603, 5);
1 row created.

SQL> INSERT INTO Rating VALUES (704, 604, 4);
1 row created.

SQL> INSERT INTO Rating VALUES (705, 605, 5);
1 row created.

```

5.1 Display Rating Records

```
SQL> SELECT*FROM Rating;
```

RATING_ID	REVIEW_ID	RATING
701	601	5
702	602	4
703	603	5
704	604	4
705	605	5

6. Retrieve Product Review Details

```
SQL> SELECT
  2     c.Customer_Name,
  3     p.Product_Name,
  4     r.Review_Text,
  5     rt.Rating,
  6     r.Review_Date
  7 FROM Review r
  8 JOIN LenskartCustomer c
  9 ON r.Customer_ID = c.Customer_ID
 10 JOIN LenskartProduct p
 11 ON r.Product_ID = p.Product_ID
 12 JOIN Rating rt
 13 ON r.Review_ID = rt.Review_ID
 14 ORDER BY r.Review_Date;
```

```
CUSTOMER_NAME
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
    RATING REVIEW_DA
    -----
Arun Kumar
Classic Black Frame
Good quality frame and comfortable to wear
    5 24-SEP-26

CUSTOMER_NAME
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
    RATING REVIEW_DA
    -----
Priya Sharma
Blue Light Glasses
Very useful for computer and laptop use
    4 25-SEP-26
```

```
CUSTOMER_NAME
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
    RATING REVIEW_DA
    -----
Rahul Kumar
Round Metal Frame
Stylish design and good fitting
    5 26-SEP-26

CUSTOMER_NAME
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
    RATING REVIEW_DA
    -----
Divya Raj
Classic Black Sunglasses
Good sunglasses with excellent look
    4 27-SEP-26

CUSTOMER_NAME
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
    RATING REVIEW_DA
    -----
Karthik S
Daily Contact Lenses
Comfortable contact lenses and easy to use
    5 28-SEP-26
```

7. Calculate Average Product Rating

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(rt.Rating) AS Average_Rating
  5 FROM LenskartProduct p
  6 JOIN Review r
  7 ON p.Product_ID = r.Product_ID
  8 JOIN Rating rt
  9 ON r.Review_ID = rt.Review_ID
 10 GROUP BY p.Product_ID, p.Product_Name
 11 ORDER BY Average_Rating DESC;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      1
Classic Black Frame
      5

      3
Round Metal Frame
      5

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      5
Daily Contact Lenses
      5

      4
Classic Black Sunglasses

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
      4

      2
Blue Light Glasses
      4
```

8. Identify Highly Rated Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(rt.Rating) AS Average_Rating
  5 FROM LenskartProduct p
  6 JOIN Review r
  7 ON p.Product_ID = r.Product_ID
  8 JOIN Rating rt
  9 ON r.Review_ID = rt.Review_ID
 10 GROUP BY p.Product_ID, p.Product_Name
 11 HAVING AVG(rt.Rating) >= 4
 12 ORDER BY Average_Rating DESC;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
          1
Classic Black Frame
          5

          3
Round Metal Frame
          5

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
          5
Daily Contact Lenses
          5

          4
Classic Black Sunglasses

PRODUCT_ID
-----
PRODUCT_NAME
-----
AVERAGE_RATING
-----
          4

          2
Blue Light Glasses
          4
```